KEMAL KILIÇASLAN

SUPER INTELLIGENCE DEVELOPER

CONTACT

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EDUCATION

KASTAMONU UNIVERSITY FACULTY OF SCIENCE AND LITERATURE MATHEMATICS - 2022 Graduate - Bachelor's Degree

EXPERIENCE

- **Bilgi Teknolojileri ve İletişim Kurumu - AI Instructor** Seasonal (July 2024 - July 2024)**:** Delivered face-to-face AI training (Python, ML, deep learning, computer vision) to 20+ participants using a hands-on curriculum built on TensorFlow, OpenCV and GPU-accelerated model training demos, guiding students through end-to-end model development from data preparation to deployment with an 85% project completion rate.
- **CALORIN - Computer Vision Developer** Remote (Aug 2024 - Sep 2024)**:** Developed a real-time food volume estimation system using YOLOv8 and OpenCV for pixel-level segmentation of fruits and vegetables, applying geometric calculations to convert segmented areas into calorie estimates with an optimised low-latency inference pipeline validated across diverse lighting and scale conditions.
- **ULTRALYTICS - AI Content Creator & Brand Partnership** (March 2026 – July 2026)**:** Produced a sponsored LinkedIn post and a YouTube tutorial for the Ultralytics Platform launch, demonstrating the full computer vision pipeline (annotation, training, export, deployment) using YOLO26 on two original real-world datasets; content delivered in compliance with FTC guidelines and Ultralytics Brand Guidelines.

PROJECTS

- **NYC311 Service Channel and Requester Behavior Analytics System:** This project implements an end-to-end analysis of 22.7M NYC311 service requests with PostgreSQL, R, Python and Spark/Hive. It models how requesters choose between phone, web and mobile, predicts phone requests with LightGBM (AUC 0.883), monitors drift with champion/challenger retraining, and forecasts call-center demand 28 days ahead.
- **Document and Identity Fraud Detection System:** This project implements document and identity fraud detection for savings-finance applications. A field-level ResNet-18 finds forged ID fields (ROC-AUC 0.989), and rules plus LightGBM fill the analysts' review queue. Monthly monitoring flags a new device-farm fraud the rules miss, and two candidate rules raise its recall from 13.5% to 82.0%.
- **Garbage Classification with Convolutional Neural Network (CNN):** This project implements a CNN model trained on the TrashNet dataset with TensorFlow/Keras. The model learns to tell apart six types of waste (cardboard, glass, metal, paper, plastic, and trash) and reaches 62.2% accuracy. In recycling systems, it can read data from a photo in real time.
- **Facial Expression Recognition System:** This project implements an emotion recognition system that identifies one of seven emotions (anger, disgust, fear, happiness, neutrality, sadness, or surprise) using the VGG19 architecture, with transfer learning and data augmentation techniques applied on the FER-2013 dataset. This particular model achieves an accuracy of 66.33% on the dataset.
- **Vehicle Speed Estimation System:** This project implements real-time vehicle speed estimation using YOLOv8 and a perspective transform. It follows several cars at once inside a marked region, turns the geometry into a speed in km/h for each one, and saves an annotated video for traffic monitoring.
- **Safety Lane Violation Detection System:** This project implements real-time detection and monitoring of the emergency lane using YOLOv12. When a vehicle crosses the safety boundary, an alert is issued and the violation count is updated instantly. Color-coded overlays make each violator easy to identify for inspection purposes.
- **Traffic Signs Recognition, Vehicle Plate and Person Blurring System:** This project implements a real-time recognition system for 24 different regulatory traffic signs commonly encountered in vehicle traffic, using YOLOv11. To protect privacy, it instantly blurs license plates and people. Thanks to this feature, it was developed for autonomous vehicles and GDPR-compliant surveillance systems.
- **Vehicle Distance Measurement System:** This project implements real-time distance measurement with YOLOv12 for in-vehicle cameras. Using a perspective geometry method, it calculates how far away nearby vehicles are in three lanes (left, front, and right), adjusts its warning thresholds as conditions change, blurs license plates for privacy, and shows color-coded safety warnings.
- **Road Surface Obstacle and Traffic Warning Object Segmentation System:** This project implements real-time, pixel-by-pixel processing of road hazards, obstacles, and traffic warning signs from an in-vehicle camera, using instance segmentation with the YOLO26x-seg model. It enables autonomous vehicles or ADAS systems to see more clearly what is in front of them.
- **Industrial Safety Gear Detection System:** This project implements real-time PPE checks using a custom YOLO26x model trained on the Ultralytics Platform. For each worker it spots, color-coded overlays show whether the helmet, safety vest, gloves, and face mask are on. Accuracy lands at 97.1% mAP50 with 94.6% precision and recall.
- **Vehicle and Pedestrian Traffic Analysis System:** This project implements zone-aware traffic analysis using a custom model trained with YOLO26x on the VisDrone dataset and exported to ONNX format, together with BoT-SORT tracking that I wrote from scratch in C++. The output is displayed live on a dashboard that analyzes people and vehicles in the crosswalk, road, and sidewalk zones.
- **Vehicle Plates Detection and Reading System:** This project implements real-time detection with a custom-trained, transformer-based RF-DETR model and license plate reading with Fast Plate OCR. It detects license plates in traffic footage, reads them, and checks each one against the Turkish plate format, then saves the results to a labeled video for AVPR.
- **Real-Time Colonoscopy Polyp Segmentation System:** This project implements real-time polyp segmentation for colonoscopy video using a PyTorch U-Net, a frame-quality check, and an IoU tracker that alarms after three consecutive frames. Training on polyp-free frames cuts false alarms on normal colon from 30.6 to 6.2 per minute, and the model scores 0.853 Dice on data from another hospital.
- **On-Premise RAG Document Assistant System:** This project implements an on-premise RAG assistant: an open-weight LLM (Qwen2.5 on Ollama) answers Turkish questions on internal PDFs and cites pages. Semantic search runs on bge-m3 embeddings in pgvector, and role-based access control limits users to allowed documents. On TQuAD, 83.7% of answers are correct when the right chunk is retrieved.

PERSONAL INITIATIVES

- Kastamonu Üniversitesi Matematik & Bilim Topluluğu **(Founding President)** (Dec 2021 - June 2022)
- Türkiye Matematik Kulübü **(Member)** (Dec 2020 - Present)
- Niğde Ömer Halisdemir Üniversitesi – TeknoKonferans NÖHÜ'25 **(Speaker)** (May 2025)

CERTIFICATES

- Machine Learning Specialization **Stanford University & DeepLearningAI**
- Deep Learning Specialization **DeepLearningAI**
- Mastering Programming with MATLAB **Vanderbilt University**
- Version Control **Meta**
- Convolutional Neural Networks in TensorFlow **DeepLearningAI**
- Mathematics for Machine Learning Specialization **DeepLearningAI**
- Self-Driving Cars Specialization **University of Toronto**
- EY Technology Risk Virtual Job Simulation **EY**